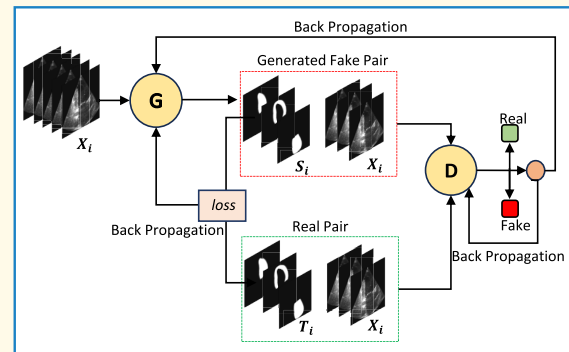


Automatic Segmentation of 2-D Echocardiography Ultrasound Images by Means of Generative Adversarial Network

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Abstract—Automated cardiac segmentation from 2-D echocardiographic images is a crucial step toward improving clinical diagnosis. Anatomical heterogeneity and inherent noise, however, present technical challenges and lower segmentation accuracy. The objective of this study is to propose a method for the automatic segmentation of the ventricular endocardium, the myocardium, and the left atrium (LA), in order to accurately determine clinical indices. Specifically, we suggest using the recently introduced pixel-to-pixel generative adversarial network (Pix2Pix GAN) model for accurate segmentation. To accomplish this, we integrate the backbone PatchGAN model for the discriminator and the UNET for the generator, for building the Pix2Pix GAN. The resulting model produces precisely segmented images because of UNET's capability for precise segmentation and PatchGAN's capability for fine-grained discrimination. For the experimental validation, we use the cardiac acquisitions for multistructure ultrasound segmentation (CAMUS) dataset, which consists of echocardiographic images from 500 patients in two-chamber (2CH) and four-chamber (4CH) views at the end-diastolic (ED) and end-systolic (ES) phases. Similar to state-of-the-art studies on the same dataset, we followed the same train–test splits. Our results demonstrate that the proposed generative adversarial network (GAN)-based technique improves segmentation performance for clinical and geometrical parameters compared with the state-of-the-art methods. More precisely, throughout the ED and ES phases, the mean Dice values for the left ventricular endocardium (LV_{endo}) reached 0.961 and 0.930 for 2CH, and 0.959 and 0.950 for 4CH, respectively. Furthermore, the average ejection fraction (EF) correlation and mean absolute error (MAE) obtained were 0.95 and 3.2 mL for 2CH, and 0.98 and 2.1 mL for 4CH, outperforming the state-of-the-art results.

Index Terms—Echocardiography, ejection fraction (EF), generative adversarial network (GAN), segmentation, ultrasound.



I. INTRODUCTION

ULTRASOUND imaging, also known as sonography, has revolutionized various fields of healthcare, providing a noninvasive and real-time visualization of internal organs and structures. It utilizes high-frequency sound waves to produce dynamic images, allowing for the examination of tissues, organs, and blood flow without the use of ionizing radiation.

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This technology has become a fundamental diagnostic tool across multiple medical specialties due to its safety, accessibility, and cost-effectiveness [1].

In the field of cardiology, ultrasound imaging has significantly advanced the understanding and management of cardiovascular diseases [2]. By utilizing echocardiography, cardiologists can visualize the chambers, valves, and surrounding structures of the heart, assessing their function and identifying the abnormalities. This noninvasive approach enables the precise evaluation of cardiac anatomy and function, aiding in the early detection and accurate diagnosis of various cardiovascular conditions [3], [4].

However, extraction of clinically relevant information from 2-D echocardiographic ultrasound images requires a careful analysis, which often relies on the interpretation of clinical indices [5], [6]. This process involves extracting vital information, guided by various clinical indices, from the images [7]. By using straightforward image processing meth-

Highlights

- This study proposes a novel method for automated segmentation of left ventricular echocardiography based on the generative model.
- We propose the utilization of Pix2Pix GAN, incorporating the PatchGAN model for the discriminator and the UNET for the generator, for cardiac segmentation.
- Results showed that our proposed technique outperformed the state-of-the-art results in terms of the estimation of clinical indices.

ods, such as segmentation and tracking, these indices can be extracted [8]. However, accurate delineation of cardiac structures, particularly, the left ventricular endocardium (LV_{endo}), during end diastole (ED) and end systole (ES), is essential for the precise extraction of parameters, such as the ejection fraction (EF) of the left ventricle (LV) [9], [10].

One of the most difficult issues in segmenting echocardiographic images is the inherent variety of heart anatomy among individuals. More specifically, the shape, size, and orientation of the heart can vary significantly. Moreover, variability is further determined by factors, such as cardiac diseases and differences in imaging protocols [11]. In this regard, several research works have been completed to address the challenges.

Traditional automatic cardiac segmentation methods [12] have played an important role in this field, providing a way to automate the process of distinguishing heart structures in echocardiographic images. These algorithms have substantial challenges due to the complexity of the heart anatomy and its changes across patients, as well as the presence of noise and abnormalities in ultrasound images [13]. In the last decade, machine learning techniques have become well-established tools for image segmentation tasks [14], e.g., random forests (RFs) [15] and support vector machines (SVMs) [16] have been successfully applied to segment cardiac structures. These approaches are, however, constrained by the requirement for feature engineering and domain-specific knowledge. Moreover, these techniques [17] often encounter several challenges that can limit their performance and effectiveness. In particular, one major issue is their struggle to handle complex and ambiguous boundaries between object classes.

In recent decades, several studies have been done on echocardiographic ultrasound image segmentation based on classical image processing techniques, such as thresholding [18], edge detection [19], and region-based methods [20]. Alternatively, due to their adaptability to various image features, intensity-based [21] and level-set approaches [22] have recently gained prominence. However, these methods often struggle with the complexities of echocardiographic images, such as noise and variability in tissue appearance. These approaches are not robust enough to handle the complex structures and fine details present in such images.

In contrast, recent advancements in deep learning (DL)-based techniques [23], primarily leveraging convolutional neural networks (CNNs) [24], have marked a significant breakthrough in echocardiographic image segmentation. Models based on architectures, such as UNET [25], [26], [27] and

fully convolutional networks (FCNs) [28], have emerged as powerful tools. For instance, Jafari et al. [28] utilized FCNs to perform echocardiography segmentation, enabling end-to-end learning and eliminating the need for manual feature extraction. Deep models, such as UNET [26], [27], [29], and its variations, such as 3-D UNET [30] and 3-D V-Net [31], have shown significant potential in cardiac image segmentation, including tasks, such as segmenting the left ventricular, right ventricular, and myocardium [32].

Nevertheless, the need for ever more accurate and robust segmentation is still evident, as even slight errors can lead to misinterpretations of clinical indices, potentially impacting patient care. To address these limitations, generative adversarial networks (GANs) [33], [34] have emerged as a valuable solution. Because of the generative capabilities stemming from the competition between a generator (G) and a discriminator (D), GAN-based approaches are, in fact, capable of achieving enhanced performance, producing more accurate, detailed, and contextually relevant segmentations, even in scenarios with complex object boundaries. In other words, GAN leverages adversarial training to facilitate data-efficient training, which makes them a promising solution to address the limitations of traditional segmentation models. Guo et al. [35] introduced a CNN architecture known as dual network GANs (DNGANs) designed specifically for precise segmentation of pediatric echocardiography images. This innovative framework incorporates a parallel structure comprising FCN and UNET to enhance network capacity. In addition, the utilization of a multiscale loss function facilitates the learning of features from both long and short distances by the generator and discriminator, further enhancing the model's effectiveness in segmentation tasks. Arafati et al. [36] utilized the cardiac acquisitions for multistructure ultrasound segmentation (CAMUS) dataset for multilabel segmentation of four-chamber (4CH) views. Employing FCNs and an adversarial training strategy with GAN, they achieved automated segmentation of all 4CHs with notable results. Specifically, the Dice metric scores were 92.1%, 86.3%, 89.6%, and 91.4% for the LV, right ventricle (RV), left atrium (LA), and right atrium (RA), respectively. Moreover, the correlation between automatic and manual segmentation for LV volumes was 0.94 and 0.93 for the end-diastolic volume and the end-systolic volume, respectively.

In the state of the art, several methods have been proposed for automated segmentation in 2-D echocardiography. Oktay et al. [37] proposed a DL model based on anatom-

ically constrained neural networks (ACNNs), which can be considered as a generic training strategy that incorporates anatomical prior knowledge into CNNs. Their approach was based on a new regularization model that offers improved cardiac image enhancement and segmentation. The performance evaluation of their method was conducted using the challenge on endocardial three-dimensional ultrasound segmentation (CETUS) dataset, yielding notable results for the segmentation of the 3-D LV_{endo} structure. More specifically, the average Dice values of LV_{endo} for ED and ES were found to be 0.936 and 0.913, respectively. On the other hand, clinical parameters based on average EF and mean absolute error (MAE) were found to be 0.807 and 5.5 mL, respectively. However, ACNNs' reliance on anatomical constraints may limit adaptability to novel anatomical variations, leading to a trade-off between accuracy and efficiency when dealing with novel anatomies.

Leclerc et al. [38] presented an application of a DL technique based on UNET for accurate segmentation of cardiac structures in 2-D echocardiography images taken from the CAMUS dataset [38] (the dataset is available at <https://www.creatis.insa-lyon.fr/Challenge/camus/link>). Their approach reached an EF correlation (corre) and an MAE as 0.845 and 5.0 mL, respectively. However, the Dice coefficients for ED and ES (0.936 and 0.912) were the same as the ones achieved in the analysis of LV_{endo} reported in the previously mentioned study [37]. To further improve echocardiography segmentation, Sfakianakis et al. [39] introduced an ensemble model merging CNN and UNET architectures for refined echocardiography segmentation, targeting specific cardiac regions in the CAMUS dataset. Their proposed method, called geometrically constrained ultrasound data augmentation in UNET (GUDU), integrated anatomical constraints into the enhanced Dice loss function, achieving the Dice coefficients of 0.946 and 0.929 for the LV_{endo} analysis at ED and ES stages. However, the average EF corre and MAE were measured at 0.897 and 4.0 mL, respectively.

To improve the average EF corre and MAE, Wei et al. [40] introduce two colearning strategies, namely, appearance-level colearning (CLA) and shape-level colearning (CLS), which improved temporal consistency in segmentation sequences from ultrasonic cardiac data. Their approach reached an average EF corre and an MAE value of 0.926 and 4.0 mL, respectively, with an improvement of up to 3% of EF value compared with the results from [39].

One of the main key challenges yet to be addressed in the current state of the art on echocardiography is the large disagreement between the results of clinical parameters determined by the clinicians and the geometrical parameters determined by the machine learning-based models. To tackle this challenge, here, we utilize GAN, which are considered a powerful tool for segmentation purposes, as they are able to extract complex information from images and produce segmentations that are highly realistic and contextually coherent to the original data. Specifically, we present the case of the automated segmentation of 2-D echocardiography images (taken from the CAMUS dataset) based on a pixel-to-pixel GAN (Pix2Pix GAN) and report the segmentation performance on the LV_{endo} , the left ventricular myocardium (LV_{myo}),

and the LA (LV_{atr}). To the best of our knowledge, this work is the first one to apply Pix2Pix GAN to echocardiography segmentation.

This article is structured as follows. Section II presents the materials and methods, which consist of the dataset, the proposed approach, the training strategy, and the evaluation metrics. Section III provides the results. Finally, Sections IV and V, respectively, report the discussion and conclusion of this research.

II. MATERIALS AND METHODS

A. Dataset

To address the challenges related to the LV echocardiogram, we performed experiments on the CAMUS dataset, consisting of echocardiographic sequences from 500 patients (450 train sets and 50 test sets). We also followed the same train-test split mentioned in the CAMUS challenge [38]. The dataset includes patients with a variety of cardiac conditions. It focuses on 2-D echocardiography and specifically includes two-chamber (2CH) and 4CH view sequences for both ED and ES. These imaging modalities capture the cardiac motion and structure from different angles, providing comprehensive information about the heart. The ground truth consists of the annotations for both the ED and ES frames, in both 2CH and 4CH perspectives, based on the LV_{endo} , the LV_{myo} , and the LA. GE Vivid E95 ultrasound scanners (GE Vingmed Ultrasound, Horten, Norway) and a GE M5S probe (GE Healthcare, Chicago, IL, USA) were used to collect the entire dataset. A detailed explanation of the dataset is given in [38].

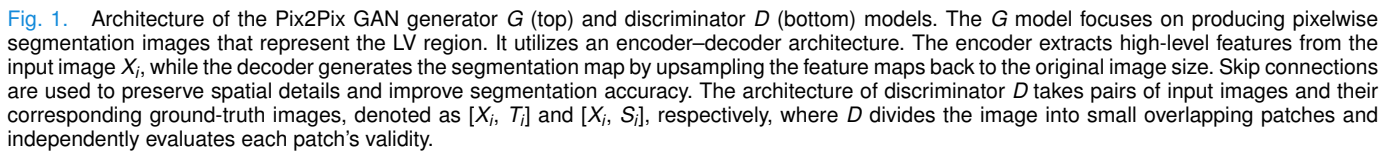
B. GAN Model

In this study, we utilize the Pix2Pix GAN [41], which is a powerful approach for medical image segmentation due to its unique characteristics. Unlike traditional segmentation models, such as UNET [42], [43], Pix2Pix GAN can generate pixel-level outputs, providing more fine-grained segmentation results. It captures global context, ensuring a better overall structure of generated data. Moreover, Pix2Pix GAN can generate diverse and creative outputs. In comparison with traditional GAN [33], Pix2Pix GAN enforces direct supervision, allowing for more precise and sharper segmentation. Its adversarial training framework encourages the model to learn and produce visually realistic and accurate semantic (one single object at a time) segmentations, making it a compelling choice for segmentation tasks.

The architecture consists of two main components, G and D . These two networks are trained simultaneously in an adversarial manner, where G tries to produce output images (segmented) that match the ground-truth images (T_i), while D is concerned with distinguishing between the segmented images (S_i) and the ground truth T_i .

1) *Generator*: The G architecture follows a UNET structure. It takes an input image (X_i) and transforms it into an output image S_i . G typically follows an encoder-decoder architecture with skip connections; see Fig. 1 (top).

The encoder path of the UNET is responsible for capturing the context and extracting features from the input image.



The decoder path of the UNET takes the high-level features from the encoder path and gradually upsamples them to generate the output image. It consists of a series of upsampling layers followed by convolutional layers. The upsampling layers increase the spatial dimensions of the feature maps, while the convolutional layers decrease the number of feature channels. Skip connections are introduced to fuse the low-level features from the encoder path with the upsampled features.

real and generated patches. This discriminator employs a series of convolutional layers to process the image patches. These layers apply a set of filters to extract features from the input, acting as downsampling layers. After each convolutional layer, a leaky ReLU activation function is applied to introduce nonlinearity into the network. The purpose of the activation function is to reduce the spatial dimensions of the feature maps while increasing their depth across the downsampling layers. Additional convolutional layers are typically applied after downsampling to capture more complex features. Once all patches have been evaluated, the discriminator provides feedback to the generator, indicating the realism of its generated patches. The generator then adjusts its parameters through backpropagation, aiming to produce more realistic image patches in subsequent iterations. The detailed information about the layers and the architecture of G and D is illustrated in Fig. 1.

During training, the weights of G and D are optimized using backpropagation and stochastic gradient descent. The learning

process involves alternating updates of the G and D networks. In our case, G utilizes two loss functions during the training, namely, binary cross-entropy loss (L_{bce}) and pixelwise loss (P_{loss}). L_{bce} is used to calculate the adversarial loss between the segmented images and the discriminator's predictions. In other words, it is used to train G to produce a segmented image S_i that D cannot distinguish from T_i

$$L_{bce} = -\frac{1}{N_s} \sum_{i=1}^{N_s} [l_r \times \log(D(T_i)) + (1 - l_r) \times \log(1 - D(S_i))] \quad (1)$$

where N_s is the number of samples in the batch. S_i is the segmented image; $D(T_i)$ and $D(S_i)$, respectively, indicate the D 's prediction for the real and segmented masks; and l_r is the real label. The discriminator typically produces a probability score indicating the likelihood of the input image being real (ground truth) or fake (generated by the generator). Here, l_r represents binary labels that distinguish between real (1) and fake (0) samples. Specifically, for real data samples (where $l_r = 1$), the loss term $l_r \times \log(D(T_i))$ encourages the discriminator to output a high probability for real images. The $\log(D(T_i))$ term penalizes the model if it predicts a low probability for real images. Conversely, for fake data samples (where $l_r = 0$), the loss term $(1 - l_r) \times \log(1 - D(S_i))$ encourages the discriminator to output a low probability for fake images. The $\log(1 - D(S_i))$ term penalizes the model if it predicts a high probability for fake data.

P_{loss} measures instead the similarity between S_i and T_i at the pixel level. It helps G to produce output images that are visually similar to the T_i . The P_{loss} is generally known as MAE loss and can be written as follows:

$$P_{loss} = \frac{1}{N_{pix}} \sum_{i=1}^{N_{pix}} \|T_i - S_i\| \quad (2)$$

where N_{pix} is the number of pixels in each image. $\|\cdot\|$ indicates the absolute value of the differences between T_i and S_i .

The total generator loss (G_{loss}) is then calculated by combining L_{bce} with P_{loss} , such that the G learns to generate visually realistic S_i

$$G_{loss} = L_{bce} + \lambda \cdot P_{loss} \quad (3)$$

where λ is a weighting factor that controls the influence of the pixelwise loss with respect to the adversarial loss L_{bce} . In each training iteration, a batch of paired images (where each pair contains an input X_i and a target image T_i) is selected from the dataset. G takes as input X_i and produces S_i . The weights of G are then updated by minimizing L_{bce} in order to "mislead" D .

The loss function of D also contains two components. The first one is the real discriminator loss (D_{rloss}), which helps D to classify real images correctly

$$D_{rloss} = -\sum_{i=1}^{i=N} \log(D[X_i, T_i]). \quad (4)$$

Algorithm 1 Training Process

Input: $D, G, GAN, Dataset(X_i, T_i), N_{epoch}, batch_size, n_{patch}$

Output: S_i

- 1: $Iteration \leftarrow 0$
- 2: $N_{batch} \leftarrow length(X_i)/batch_size$
- 3: **for** $k = 1; k = N_{epoch}; k++$ **do**
- 4: **for** $i = 1; i = N_{batch}; i++$ **do** $\triangleright$ Mini-batch learning
- 5: $S_i \leftarrow G(X_i, n_{patch})$ $\triangleright$ Batch of fake samples
- 6: $D_{rloss} \leftarrow CalculateRealLoss([X_i, T_i])$ $\triangleright$ Eq. (4)
- 7: $D_{floss} \leftarrow CalculateFakeLoss([X_i, S_i])$ $\triangleright$ Eq. (5)
- 8: $D_{loss} \leftarrow D_{rloss} + D_{floss}$ $\triangleright$ Eq. (6)
- 9: $Update_discriminator(D_{loss})$
- 10: $G_{loss} \leftarrow CalculateGloss(S_i, T_i)$ $\triangleright$ Eq. (3)
- 11: $Update_generator(G_{loss})$
- 12: $Iteration \leftarrow Iteration + 1$
- 13: **end for**
- 14: **end for**

The second component is the fake discriminator loss (D_{floss}), which is based on the D 's output when provided the S_i

$$D_{floss} = -\sum_{i=1}^{i=N} \log(1 - D[X_i, S_i]). \quad (5)$$

This second component of the loss function enforces D to classify segmented images correctly. The total discriminator loss (D_{loss}) is then a combination of the real and fake losses

$$D_{loss} = D_{rloss} + D_{floss}. \quad (6)$$

The G and D models are trained simultaneously by updating their weights based on their respective losses to improve the overall performance of the Pix2Pix GAN. Throughout the training process, we executed our training process for N_{tot} iterations. By closely monitoring the training process, we saved images on the minimum value of G loss to ensure that the model achieves an optimal level of performance (see Fig. 2). The training process is graphically illustrated in Fig. 3 and further detailed in Algorithm 1.

D. Evaluation Metrics

To assess the effectiveness of Pix2Pix GAN, we use five key geometrical metrics, namely, Dice, Hausdorff distance (HD), mean absolute difference (MAD), and two clinical metrics, such as corre and EF

$$Dice = \frac{(2 \times |S_i \cap T_i|)}{(|S_i| + |T_i|)} \quad (7)$$

$$MAD = \left(\frac{1}{n}\right) \times \sum |T_i - S_i| \quad (8)$$

$$HD(T_i, S_i) = \max \left(\max_{t \in T_i} \min_{s \in S_i} d(t, s), \max_{s \in S_i} \min_{t \in T_i} d(s, t) \right) \quad (9)$$

$$LV_{EF} = \frac{LV_{EDV} - LV_{ESV}}{LV_{EDV}} \quad (10)$$

$$EF \text{ corre} = \frac{\text{cov}(EF(T_i), EF(S_i))}{\sigma_{EF}(T_i) \times \sigma_{EF}(S_i)}. \quad (11)$$

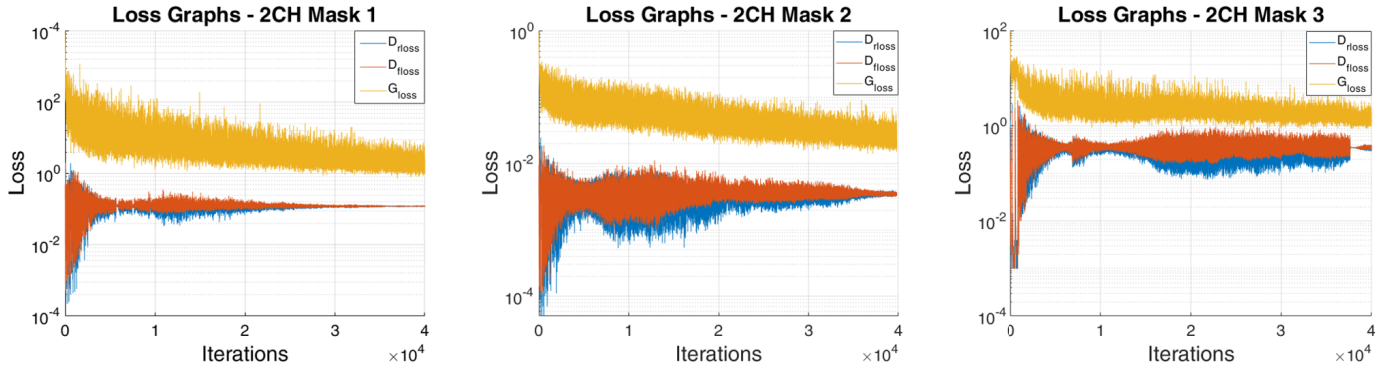


Fig. 2. Training progress of Pix2Pix GAN for LV_{endo} (left column), myocardium (central column), and LA segmentation (right column). It represents the loss trends on the 2CH dataset (see the link for the case of 4CH view). The red and blue lines represent the discriminator D_{floss} and D_{loss} losses, respectively, while the orange lines represent the generator loss (G_{loss}) during the training process. These loss curves provide insights into the learning dynamics of Pix2Pix GAN, showing how it iteratively refines its ability to generate accurate segmentations of the specified cardiac structures.

In the context of cardiac evaluation, Dice, HD, and MAD, are employed to evaluate the accuracy of a segmentation model's performance when comparing predicted segmentations S_i with ground-truth segmentations T_i . These metrics provide quantitative measures of the extent of overlap and alignment between the predicted and real anatomical regions of interest. Specifically, Dice assesses spatial overlap (7), MAD quantifies the average pixel value differences (8), HD calculates the maximum distance between two coordinates [see (9)], and $d(t, s)$ represents the distance metric between point t in T_i and point s in S_i . For cardiac health assessment, the left ventricular EF (10) is computed by comparing the segmented left ventricular end-diastolic volume (LV_{EDV}) and the left ventricular end-systolic volume (LV_{ESV}). EF corre is computed in (11) by calculating the covariance (cov) and standard deviation (σ) value between the two arrays, EF of S_i and T_i . These metrics collectively provide a comprehensive evaluation of the accuracy and clinical relevance of the segmentation model's output, aiding in the assessment of cardiac function and disease diagnosis.

III. RESULTS

In the experimental evaluation, we utilized the proposed approach based on the Pix2Pix GAN model, aiming to generate segmented results for both 2CH and 4CH views. To this end, the Pix2Pix GAN model underwent training using the CAMUS dataset for real-time prediction purposes. During the evaluation phase, a set of test images (Y_i) served as input to our previously trained GAN model. The model's primary task was to predict segmented regions corresponding to LV_{endo}, LV_{myo}, and L_{atr} . The training progression of Pix2Pix GAN is illustrated in Fig. 2. Table I provides an overview of the parameters employed during the training process. During training, images are split into smaller patches, and patch size is 70×70 to improve the model's understanding of local details and spatial relationships. In our mini-batch learning strategy, using a batch size of 1, each training iteration handles the process of calculating and backpropagating gradients on image patches. The generator creates a fake output patch based on the input, while the discriminator assesses their realism. The training process completes an epoch after processing

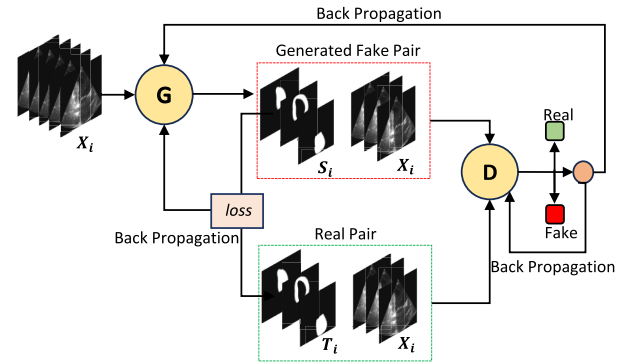


Fig. 3. Training alternates between the generator (G) and the discriminator (D). G aims to create fake segmentation images S_i that resemble the ground truth T_i , making it challenging for D to distinguish them. This continues until G can consistently produce accurate segmentation images S_i that closely resemble the ground truth T_i . In each training iteration, paired input images (X_i and T_i) and (X_i and S_i) are used to update the discriminator D .

all images in the training set. We continued training for 1000 epochs to ensure that we achieved the lowest possible loss value. A learning rate of 0.0002 was selected to promote training stability and minimize oscillations.

Some examples of qualitative segmentation results utilizing the proposed Pix2Pix GAN are shown in Fig. 4. Following the successful generation of segmented images with clearly defined and accurate boundaries (by visual inspection), we proceeded to conduct a thorough analysis aimed at numerically evaluating the performance of the proposed technique with respect to the geometrical parameters introduced in Section II-D. Moreover, we compared the performance of Pix2Pix GAN with other augmentation techniques from the literature. The qualitative and quantitative comparative results are reported in Fig. 5 and Table II, respectively. The results stemming from this comparative analysis yielded valuable insights into the strengths and weaknesses of each method.

In addition, we provided the clinically relevant parameters based on the segmented images in both the ED and ES phases. More specifically, after calculating the volume in these phases, we calculated the EF based on (10). The statistical results (corre and MAE) are reported in Table III.

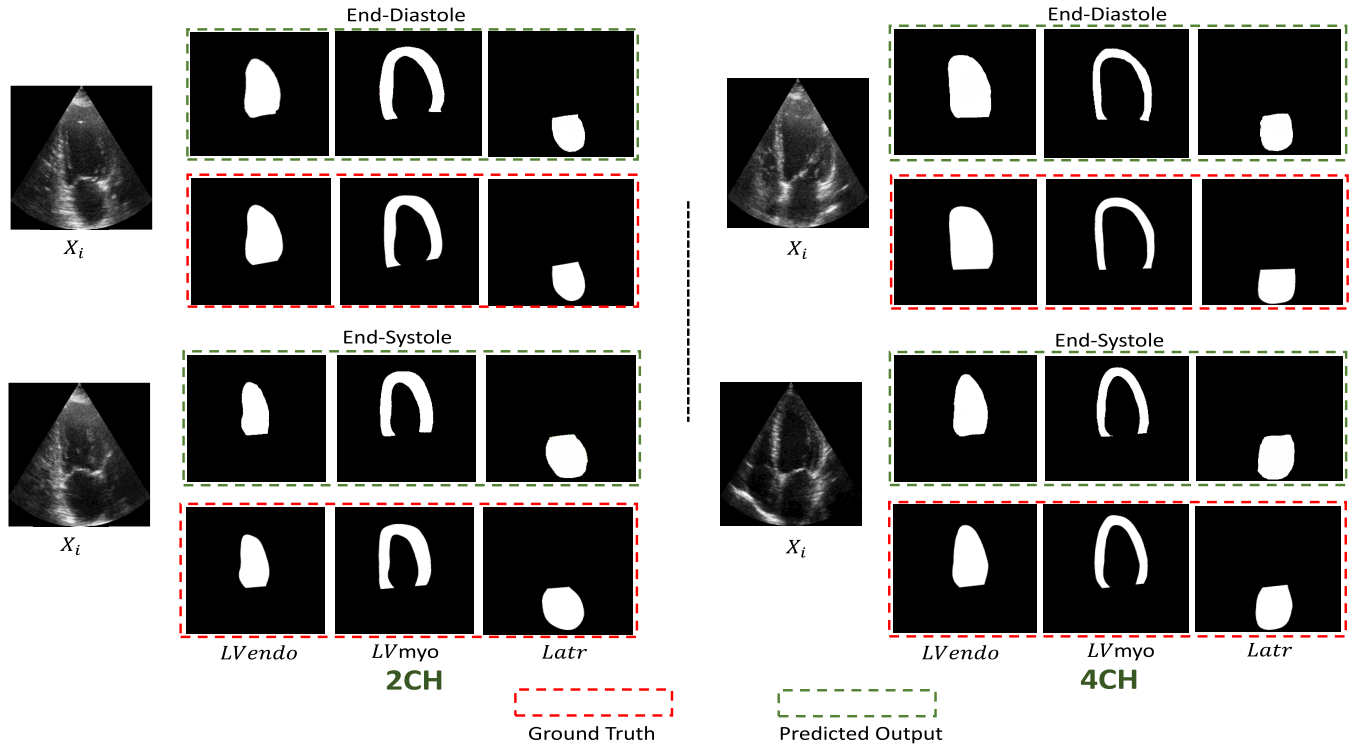


Fig. 4. Example of segmented images using the proposed Pix2Pix GAN, compared with the ground truth. The red dotted lines indicate the ground truth (T_i), while the green lines indicate the segmented results (S_i). The results are shown for the CAMUS dataset.

TABLE I

HYPERPARAMETERS FOR THE PIX2PIX GAN TRAINING PROCESS

Parameters	Value
Batch Size	1
Number of Epochs (N_{epoch})	1000
Optimizer	Adam
Learning Rate	0.0002
Loss Function	G_{loss} and D_{loss} [see Eq. (3) and Eq. (6)]

It is important to evaluate the robustness of our approach with respect to the training size. In this regard, we analyzed different parameters for both 2CH and 4CH views with different training sizes. The impact of different training sizes is shown by the boxplots reported in Figs. 6 and 7 (where, e.g., 300 training size means 150 ED and 150 ES images used for training). Fig. 8 provides the worst-performing segmentation results of our models selected from the outliers in boxplots of Fig. 6.

The comparison of our model with state-of-the-art models on the CAMUS dataset, including ACNN [37], UNET [38], CLAS [40], NN-UNET [46], and GUDU [39] networks, is given in Table IV. The computational analysis of our approach in comparison with the augmentation techniques from the literature is summarized in Table V. The generalization capability of our approach on EchoNet dataset is also analyzed; see the evaluations reported in Table VI, and Figs. 9 and 10.

The entire experimentation was carried out on a personal computer using the Anaconda Spyder tool. The experiments

utilized the TensorFlow Python API, which is compatible with Python 3.7, and employed Keras as the DL API. The hardware setup comprised an NVIDIA GeForce RTX 3060 GPU, CUDA 10.1, a 16-core processor, and 64 GB of RAM.

IV. DISCUSSION

The results obtained in our study underscore the effectiveness and robustness of our trained Pix2Pix GAN model in carrying out the segmentation task. The model consistently generated accurate segmented outcomes for the LV_{endo} , LV_{myo} , and L_{atr} regions in both 2CH and 4CH datasets, as illustrated in Fig. 4. These segmentation results were characterized by high Dice values, effectively capturing the LV area and ensuring dependable measurements of its size and function (see Table II). Importantly, the segmentation process was comprehensive, covering the entire region of interest with well-defined boundaries, leaving no gaps or areas overlooked. This complete segmentation approach facilitated a thorough analysis of the LV, enabling precise evaluation of the EF.

To enhance the model's capacity to handle diverse data and improve its adaptability across various scenarios, we conducted separate training sessions for the 2CH and 4CH datasets, focusing on LV_{endo} , LV_{myo} , and L_{atr} . It should be noted that we have developed two separate models for the two cases of 2CH and 4CH views. This setup allows us to switch between them and select the corresponding model based on the view of the recorded data (i.e., 2CH or 4CH). The outcomes of the G and D models in terms of losses are calculated throughout the training process, as visually represented in Fig. 2.

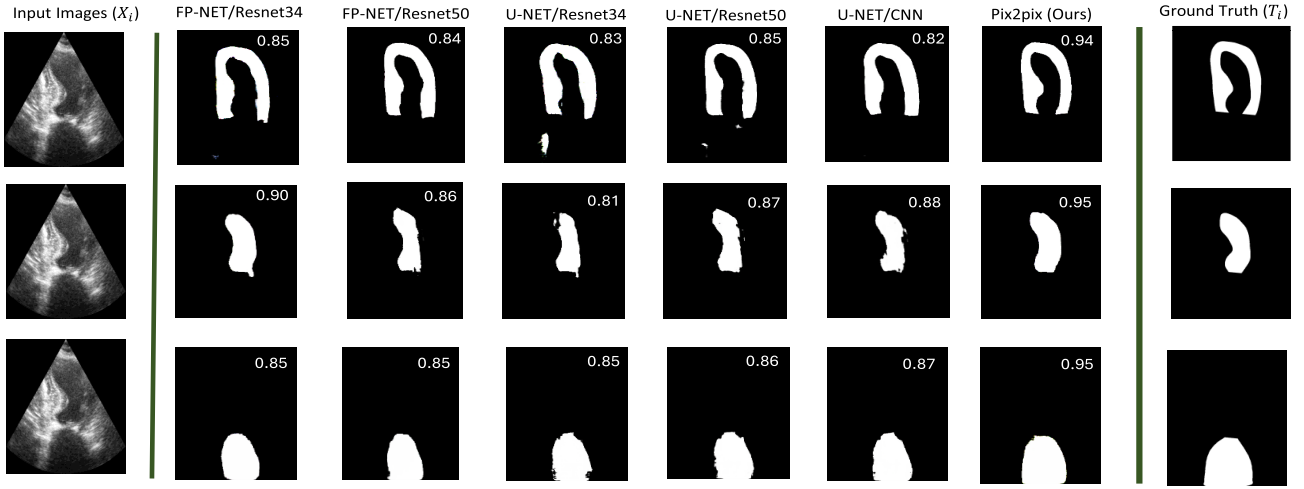


Fig. 5. Qualitative comparison between the segmentation methods from the literature and the proposed Pix2Pix GAN-based approach. The segmented results illustrate the endocardium (first row), the myocardium (second row), and the LA (third row). The given example indicates one of the most complex cardiac structures. The corresponding Dice values are overlaid to each predicted mask. The results are shown for the CAMUS dataset.

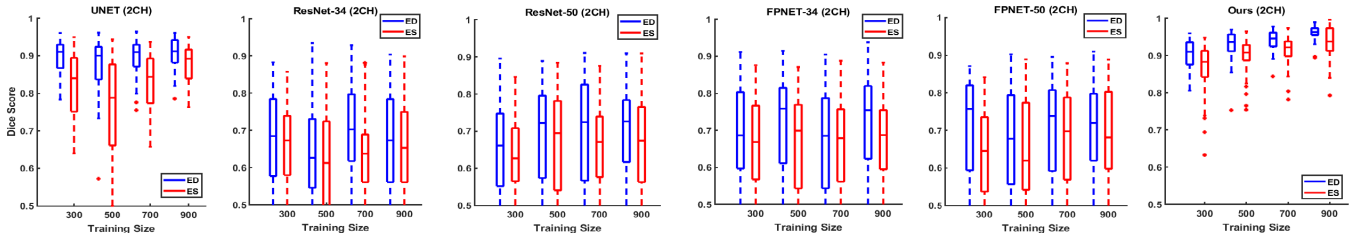


Fig. 6. Dice scores across different training sizes for the analysis of end-diastole (ED) and end-systole (ES) data in 2CH view (see the link for the case of 4CH view), comparing our proposed approach with segmentation techniques from the literature, including UNET, UNET Resnet34, UNET Resnet50, FPNET Resnet34, and FPNET Resnet50. The results are shown for the CAMUS dataset.

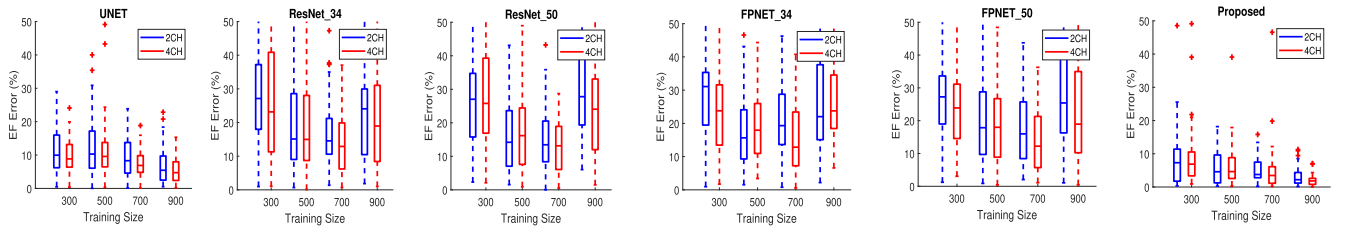


Fig. 7. Analysis of EF (%) errors for segmented images compared with ground-truth images in both 2CH and 4CH views, with varying training sizes, in comparison with UNET, UNET Resnet34, UNET Resnet50, FPNET Resnet34, and FPNET Resnet50. The results are shown for the CAMUS dataset.

As illustrated in Figs. 4 and 5, by employing a combination of adversarial training and pixelwise loss functions, our network has acquired the capability to generate sharp and highly accurate segmented images that closely resemble the ground truth. These trained weights were subsequently applied to predict test images, yielding remarkably precise predictions, as evident in Fig. 4. One noteworthy improvement is the visibly sharper edges of the segmented regions, which significantly enhances the overall quality of the segmentation's and facilitates reliable delineation of the LV area. This exemplifies the efficacy of the Pix2Pix GAN in accurately segmenting and highlighting specific regions within the LV.

A. Analysis of Geometrical Parameters

We conducted a comprehensive evaluation of our model's performance employing various geometrical metrics. In addition, we engaged in a fair comparison with other classic

models that we implemented for segmentation, as outlined in Table II. More specifically, as an example, our proposed model achieved the highest performance results of {Dice} as {0.961, 0.930} and {0.959, 0.950}, considering the ED and ES phase 2CH and 4CH for LV_{endo}, respectively.

Similarly, considering the ED and ES phases for 2CH and 4CH of LV_{endo}, the combined training provided the Dice coefficients of {0.961, 0.946}, as reported in the comparative analysis shown in Table IV. These results underscore our model's consistency and high-quality performance confirming its effectiveness in the context of cardiac image segmentation, as exemplified in Fig. 5.

B. Analysis of Clinical Parameters

We have undertaken a comprehensive examination of our model's performance concerning LV_{endo} in comparison with other widely used methods mentioned in Table III. In this

TABLE II

COMPARATIVE ANALYSIS OF GEOMETRICAL INDICES ON THE CAMUS DATASET. THE MEAN AND STANDARD DEVIATIONS (MEAN $\pm$ STD) ARE CALCULATED BASED ON RUNNING THE EXPERIMENTS FIVE TIMES

Method	Backbone	Dice _{ED}	Dice _{ES}	HD _{ED}	HD _{ES}	MAD _{ED}	MAD _{ES}
2CH-LV _{endo}							
UNET	CNN	0.91 $\pm$ 0.1	0.88 $\pm$ 0.1	2.9 $\pm$ 1.3	2.6 $\pm$ 1.6	4.8 $\pm$ 2.5	3.1 $\pm$ 2.0
UNET	Resnet34	0.66 $\pm$ 0.1	0.64 $\pm$ 0.2	9.2 $\pm$ 5.1	7.6 $\pm$ 4.1	13.4 $\pm$ 8.7	7.5 $\pm$ 6.5
UNET	Resnet50	0.70 $\pm$ 0.2	0.66 $\pm$ 0.1	8.6 $\pm$ 4.9	6.7 $\pm$ 3.0	12.7 $\pm$ 8.8	6.1 $\pm$ 6.0
FPNET	Resnet34	0.71 $\pm$ 0.1	0.66 $\pm$ 0.1	8.6 $\pm$ 5.6	6.8 $\pm$ 3.7	12.7 $\pm$ 9.2	6.7 $\pm$ 5.9
FPNET	Resnet50	0.68 $\pm$ 0.2	0.68 $\pm$ 0.1	9.5 $\pm$ 6.5	7.1 $\pm$ 4.6	13.6 $\pm$ 9.4	6.8 $\pm$ 6.4
FCN-GAN	FCN	0.88 $\pm$ 0.1	0.81 $\pm$ 0.1	18.3 $\pm$ 1.5	19.7 $\pm$ 1.8	4.7 $\pm$ 3.1	3.35 $\pm$ 2.2
DNGAN	VGG	0.87 $\pm$ 0.1	0.84 $\pm$ 0.1	11.5 $\pm$ 4.1	12.8 $\pm$ 4.1	6.6 $\pm$ 5.0	4.2 $\pm$ 3.0
Pix2Pix GAN	DCNN	0.96 $\pm$ 0.01	0.93 $\pm$ 0.02	1.6 $\pm$ 0.2	5.3 $\pm$ 7.1	1.9 $\pm$ 0.5	1.7 $\pm$ 0.5
2CH-LV _{myo}							
UNET	CNN	0.53 $\pm$ 0.1	0.58 $\pm$ 0.1	12.8 $\pm$ 7.6	10.5 $\pm$ 5.9	18.2 $\pm$ 10.4	15.8 $\pm$ 8.8
UNET	Resnet34	0.54 $\pm$ 0.1	0.60 $\pm$ 0.1	11.0 $\pm$ 7.4	8.5 $\pm$ 5.0	16.8 $\pm$ 9.6	14.3 $\pm$ 7.8
UNET	Resnet50	0.57 $\pm$ 0.2	0.58 $\pm$ 0.1	9.1 $\pm$ 5.5	8.6 $\pm$ 4.2	15.5 $\pm$ 9.0	14.9 $\pm$ 7.9
FPNET	Resnet34	0.57 $\pm$ 0.2	0.58 $\pm$ 0.1	9.5 $\pm$ 5.2	8.8 $\pm$ 5.3	16.2 $\pm$ 9.6	15.3 $\pm$ 8.3
FPNET	Resnet50	0.54 $\pm$ 0.1	0.56 $\pm$ 0.2	10.6 $\pm$ 6.7	8.7 $\pm$ 4.8	17.5 $\pm$ 10.1	16.0 $\pm$ 8.5
FCN-GAN	FCN	0.73 $\pm$ 0.1	0.74 $\pm$ 0.1	18.6 $\pm$ 2.3	19.9 $\pm$ 2.4	13.8 $\pm$ 8.3	13.6 $\pm$ 6.8
DNGAN	VGG	0.74 $\pm$ 0.1	0.76 $\pm$ 0.1	10.6 $\pm$ 3.3	11.8 $\pm$ 3.8	12.0 $\pm$ 6.8	10.3 $\pm$ 5.7
Pix2Pix GAN	DCNN	0.93 $\pm$ 0.01	0.94 $\pm$ 0.01	3.1 $\pm$ 3.4	3.2 $\pm$ 3.4	2.7 $\pm$ 1.2	2.2 $\pm$ 0.8
2CH-LV _{atr}							
UNET	CNN	0.46 $\pm$ 0.2	0.47 $\pm$ 0.1	8.9 $\pm$ 3.9	10.6 $\pm$ 3.2	7.2 $\pm$ 4.1	10.3 $\pm$ 3.8
UNET	Resnet34	0.47 $\pm$ 0.3	0.42 $\pm$ 0.2	13.3 $\pm$ 9.6	12.0 $\pm$ 5.1	6.8 $\pm$ 4.1	11.0 $\pm$ 4.8
UNET	Resnet50	0.51 $\pm$ 0.2	0.46 $\pm$ 0.2	8.2 $\pm$ 4.6	10.3 $\pm$ 4.3	6.5 $\pm$ 3.8	10.2 $\pm$ 4.3
FPNET	Resnet34	0.50 $\pm$ 0.2	0.47 $\pm$ 0.2	8.5 $\pm$ 4.8	10.6 $\pm$ 4.8	6.9 $\pm$ 3.9	10.2 $\pm$ 4.6
FPNET	Resnet50	0.48 $\pm$ 0.2	0.45 $\pm$ 0.2	8.9 $\pm$ 5.3	11.1 $\pm$ 4.8	7.1 $\pm$ 4.5	10.4 $\pm$ 3.9
FCN-GAN	FCN	0.77 $\pm$ 0.2	0.82 $\pm$ 0.2	20.7 $\pm$ 11.4	13.4 $\pm$ 10.3	3.8 $\pm$ 2.9	4.5 $\pm$ 3.0
DNGAN	VGG	0.70 $\pm$ 0.2	0.76 $\pm$ 0.17	9.7 $\pm$ 7.9	9.4 $\pm$ 7.2	5.5 $\pm$ 4.1	6.4 $\pm$ 4.1
Pix2Pix GAN	DCNN	0.92 $\pm$ 0.04	0.96 $\pm$ 0.01	3.9 $\pm$ 5.3	2.3 $\pm$ 3.9	1.1 $\pm$ 0.6	0.9 $\pm$ 0.3
4CH-LV _{endo}							
UNET	CNN	0.92 $\pm$ 0.1	0.90 $\pm$ 0.1	2.3 $\pm$ 1.1	2.6 $\pm$ 2.2	4.3 $\pm$ 2.2	2.8 $\pm$ 1.9
UNET	Resnet34	0.73 $\pm$ 0.1	0.67 $\pm$ 0.2	8.2 $\pm$ 6.1	6.8 $\pm$ 4.6	11.7 $\pm$ 9.3	6.9 $\pm$ 6.1
UNET	Resnet50	0.76 $\pm$ 0.1	0.69 $\pm$ 0.1	6.7 $\pm$ 4.3	5.5 $\pm$ 3.4	9.9 $\pm$ 7.9	5.4 $\pm$ 5.3
FPNET	Resnet34	0.74 $\pm$ 0.1	0.71 $\pm$ 0.1	8.1 $\pm$ 5.5	5.5 $\pm$ 3.7	11.6 $\pm$ 8.8	5.7 $\pm$ 5.6
FPNET	Resnet50	0.73 $\pm$ 0.2	0.71 $\pm$ 0.1	9.0 $\pm$ 6.6	6.4 $\pm$ 5.0	12.4 $\pm$ 9.6	6.6 $\pm$ 6.2
FCN-GAN	FCN	0.90 $\pm$ 0.1	0.83 $\pm$ 0.1	19.7 $\pm$ 1.5	21.0 $\pm$ 1.7	4.3 $\pm$ 2.9	3.6 $\pm$ 2.5
DNGAN	VGG	0.88 $\pm$ 0.1	0.85 $\pm$ 0.1	11.8 $\pm$ 4.0	13.0 $\pm$ 4.6	6.1 $\pm$ 3.7	4.1 $\pm$ 2.2
Pix2Pix GAN	DCNN	0.96 $\pm$ 0.01	0.95 $\pm$ 0.01	2.0 $\pm$ 2.9	5.0 $\pm$ 6.5	2.1 $\pm$ 0.9	1.6 $\pm$ 0.5
4CH-LV _{myo}							
UNET	CNN	0.58 $\pm$ 0.1	0.56 $\pm$ 0.1	12.7 $\pm$ 7.6	9.7 $\pm$ 5.7	12.2 $\pm$ 6.4	11.9 $\pm$ 5.3
UNET	Resnet34	0.55 $\pm$ 0.1	0.55 $\pm$ 0.1	10.3 $\pm$ 6.6	7.8 $\pm$ 5.1	12.8 $\pm$ 6.5	11.7 $\pm$ 5.0
UNET	Resnet50	0.56 $\pm$ 0.1	0.54 $\pm$ 0.1	9.8 $\pm$ 6.3	8.6 $\pm$ 4.9	12.2 $\pm$ 5.8	12.2 $\pm$ 4.8
FPNET	Resnet34	0.56 $\pm$ 0.1	0.54 $\pm$ 0.1	9.6 $\pm$ 6.1	7.3 $\pm$ 3.6	12.4 $\pm$ 5.7	12.5 $\pm$ 5.0
FPNET	Resnet50	0.54 $\pm$ 0.1	0.54 $\pm$ 0.1	10.8 $\pm$ 7.0	7.9 $\pm$ 4.5	13.3 $\pm$ 6.4	12.6 $\pm$ 4.8
FCN-GAN	FCN	0.73 $\pm$ 0.1	0.72 $\pm$ 0.1	19.8 $\pm$ 2.1	20.9 $\pm$ 2.5	10.3 $\pm$ 5.1	10.9 $\pm$ 4.3
DNGAN	VGG	0.77 $\pm$ 0.1	0.75 $\pm$ 0.1	9.7 $\pm$ 3.8	10.4 $\pm$ 3.5	7.4 $\pm$ 3.7	7.3 $\pm$ 3.2
Pix2Pix GAN	DCNN	0.94 $\pm$ 0.01	0.94 $\pm$ 0.01	1.5 $\pm$ 2.0	2.9 $\pm$ 3.5	1.4 $\pm$ 0.4	1.8 $\pm$ 0.5
4CH-LV _{atr}							
UNET	CNN	0.57 $\pm$ 0.2	0.58 $\pm$ 0.1	7.1 $\pm$ 2.9	8.6 $\pm$ 2.5	5.8 $\pm$ 2.7	8.7 $\pm$ 2.4
UNET	Resnet34	0.60 $\pm$ 0.22	0.59 $\pm$ 0.2	6.6 $\pm$ 3.4	8.2 $\pm$ 3.8	4.9 $\pm$ 3.4	7.9 $\pm$ 3.7
UNET	Resnet50	0.60 $\pm$ 0.2	0.63 $\pm$ 0.2	6.5 $\pm$ 3.2	7.3 $\pm$ 3.0	4.9 $\pm$ 3.4	7.5 $\pm$ 3.1
FPNET	Resnet34	0.63 $\pm$ 0.2	0.62 $\pm$ 0.1	6.1 $\pm$ 3.1	7.8 $\pm$ 3.0	4.7 $\pm$ 2.7	7.6 $\pm$ 2.7
FPNET	Resnet50	0.63 $\pm$ 0.2	0.59 $\pm$ 0.2	6.4 $\pm$ 3.1	8.3 $\pm$ 3.7	5.2 $\pm$ 3.1	8.3 $\pm$ 3.1
FCN-GAN	FCN	0.80 $\pm$ 0.2	0.88 $\pm$ 0.1	19.7 $\pm$ 11.1	11.4 $\pm$ 9.0	3.4 $\pm$ 2.2	3.5 $\pm$ 1.7
DNGAN	VGG	0.74 $\pm$ 0.2	0.84 $\pm$ 0.1	8.9 $\pm$ 6.8	7.8 $\pm$ 6.7	4.5 $\pm$ 2.3	4.9 $\pm$ 2.4
Pix2Pix GAN	DCNN	0.90 $\pm$ 0.1	0.92 $\pm$ 0.05	4.8 $\pm$ 7.4	3.6 $\pm$ 5.6	1.7 $\pm$ 1.1	1.8 $\pm$ 1.2

analysis, we have considered the essential clinical parameters relevant to the assessment of LV_{endo}. These clinical parameters encompass the determination of corre and observation of the MAE concerning LV_{EDV}, LV_{ESV}, and left ventricular EF (LV_{EF}) for 2CH and 4CH data. The examination of corre is particularly valuable, as it provides us with crucial insights into the strength and direction of the linear relationship between the T_i and the S_i data of LV_{endo}. By evaluating this corre coefficient, we can gain a better understanding of how well

our model aligns with real-world clinical measurements and identify any potential patterns or discrepancies. This analysis enables us to assess the accuracy and reliability of our model's predictions compared with other models typical used for segmentation task, ultimately guiding us in making informed decisions about the effectiveness and applicability of our model for LV_{endo} analysis. According to this analysis, the {corre and MAE} of LV_{EDV} reached {0.980, 2.2 mL} and {0.983, 1.9 mL} for 2CH and 4CH, respectively. Similarly,

TABLE III

CLINICAL EVALUATIONS ON THE CAMUS DATASET FOR THE LV_{ENDO} SEGMENTATION

	corre			MAE		
	LV_{EDV}	LV_{ESV}	LV_{EF}	$LV_{EDV}(ml)$	$LV_{ESV}(ml)$	$LV_{EF}(\%)$
UNET	0.94	0.95	0.87	10.1	6.9	6.4
UNET-Resnet34	0.55	0.67	0.42	29.9	17.3	22.2
UNET-Resnet50	0.61	0.67	0.43	26.5	19.2	26.7
FPNET-Resnet34	0.62	0.61	0.48	27.7	17.6	26.8
FPNET-Resnet50	0.62	0.66	0.38	31.1	15.2	24.4
FCN-GAN	0.90	0.94	0.80	11.5	9.8	8.2
DNGAN	0.92	0.95	0.87	12.8	7.0	6.9
Pix2Pix GAN (2CH-trained only)	0.98	0.97	0.95	4.4	3.6	3.2
Pix2Pix GAN (4CH-trained-only)	0.98	0.99	0.98	3.9	1.9	2.1
Pix2Pix GAN (trained on both 2CH and 4CH)	0.99	0.99	0.98	4.4	2.8	1.6

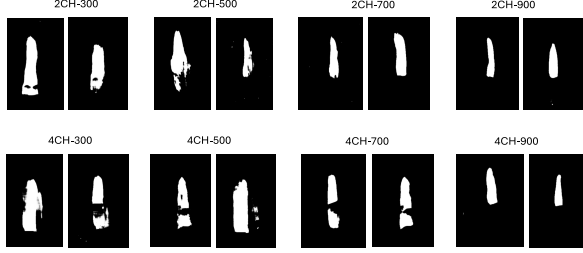


Fig. 8. Two examples (ED-ES pairs) of worst-performing segmentation results of our proposed model with different training sizes on the CAMUS dataset.

in the case of LV_{ESV} , the obtained values were {0.977, 1.9 mL} and {0.992, 1.7 mL}, respectively, for 2CH and 4CH. However, our model was also trained on both chambers, as mentioned in Table III, revealing that the {corre and MAE} of LV_{EDV} , LV_{ESV} , and LV_{EF} achieved the values of {0.99, 4.4 mL}, {0.99, 2.8 mL}, and {0.98, 1.6 mL}, respectively, which outperformed the state-of-the-art results. These results demonstrate that our approach achieves the highest corre and minimum MAE among all the methods. This could be explained by the fact that Pix2Pix GAN learns complex patterns and features within the data. In other words, our model not only provides the local features of individual pixels (information in the same patch) but also the global context (combination of all patches) for the entire image. This contextual representation capability enables our approach to make more informed decisions when distinguishing between objects and their background, leading to more precise segmentation. Unlike traditional segmentation methods that rely on predefined rules and handcrafted features, our approach Pix2Pix GAN adapts and optimizes its segmentation strategy based on the specific characteristics of the input images. This adaptability leads to more accurate and robust segmentation results, especially in cases where object boundaries are intricate or not well defined.

C. Sensitivity of Performance to Training Size

To evaluate the impact of different training data sizes, we divided our training dataset into subsets of 300, 500, 700, and 900 samples for both 2CH and 4CH views. We then focused on assessing the geometric parameters, specifically the Dice coefficient, for both 2CH and 4CH views, and for both ED and ES, when varying the training size. The results from our analysis, reported in Fig. 6, revealed a noticeable trend of performance improvement, as we increased the size of the training dataset.

TABLE IV

COMPARATIVE ANALYSIS ON THE CAMUS DATASET FOR THE LV_{ENDO} . THE RESULTS ARE REPORTED FOR THE TRAINED MODELS ON BOTH 2CH AND 4CH DATASETS TOGETHER

Method	Dice _{ED}	Dice _{ES}	HD _{ED}	HD _{ES}	MAD _{ED}	MAD _{ES}
ACNN [37]	0.936	0.913	5.6	5.6	1.7	1.7
UNET [38]	0.936	0.912	5.3	5.5	1.7	1.7
GUDU [39]	0.946	0.929	4.7	4.7	1.4	1.4
CLAS [40]	0.947	0.929	4.6	4.6	1.4	1.4
NN-UNET [46]	0.952	0.935	4.3	4.2	1.3	1.3
Pix2Pix GAN (Ours)	0.961	0.946	0.9	1.6	1.8	1.5

	corre			MAE (ml)		
	LV_{EDV}	LV_{ESV}	LV_{EF}	LV_{EDV}	LV_{ESV}	LV_{EF}
ACNN [37]	0.928	0.954	0.807	9.7	6.9	5.5
UNET [38]	0.926	0.960	0.845	11.2	7.5	5.0
GUDU [39]	0.977	0.981	0.897	6.7	4.6	4.0
CLAS [40]	0.958	0.979	0.926	7.7	4.4	4.0
NN-UNET [46]	0.977	0.987	0.857	5.9	4.0	4.7
Pix2Pix GAN (Ours)	0.995	0.993	0.98	4.4	2.8	1.6

D. Quantitative Analysis With CAMUS State-of-the-Art Models

In addition, we took into account clinical parameters in our evaluation, with a particular emphasis on the EF error, which is computed between the S_i and T_i images. Fig. 7 shows that the EF error gradually decreases while increasing the number of samples. This represents a positive result, indicating that augmenting the training data size has enhanced the accuracy of our cardiac image segmentation approach, thus holding significant potential for clinical applications in cardiology. The enhanced performance resulting from increased training data highlights the model's adaptability in optimizing its parameters. Fig. 8 illustrates the evolution of the worst-performing segmentation samples (selected from Fig. 6), as the training size increases. This figure showcases the influence of the training size on the model's capability for segmentation.

The quantitative analysis of our proposed approach is compared with state-of-the-art models, as outlined in Table IV. This analysis highlights the thorough evaluation of LV_{ENDO} . Various models, such as ACNN [37], UNET [38], GUDU [39], CLAS [40], and NN-UNET [46], have been examined based on their correlation (corre) and MAE for parameters, including LV_{EDV} , LV_{ESV} , and EF (LV_{EF}). Notably, our Pix2Pix GAN model, trained on both chambers, demonstrates superior performance with a correlation of 0.995 for LV_{EDV} , 0.993 for LV_{ESV} , and 0.98 for LV_{EF} along with a remarkably low MAE of 4.4 mL for LV_{EDV} , 2.8 mL for LV_{ESV} , and 1.6 mL for LV_{EF} . This underscores the effectiveness of our approach in accurately assessing left ventricular parameters, showcasing promising advancements in cardiac imaging techniques.

E. Computational Analysis

In Table V, we compare the computational resources required by our method with those of the alternative approaches from the literature, focusing on two key factors, namely, the number of parameters (which is a proxy for memory consumption) and training time (which is a proxy for computing power). The table indicates that the proposed Pix2Pix GAN approach utilizes a larger number of parameters (61.40 M) where generator utilized the 54 M and discriminator utilized the 6.9 M parameters and a larger training time (138 s) due to its two-network architecture and complex

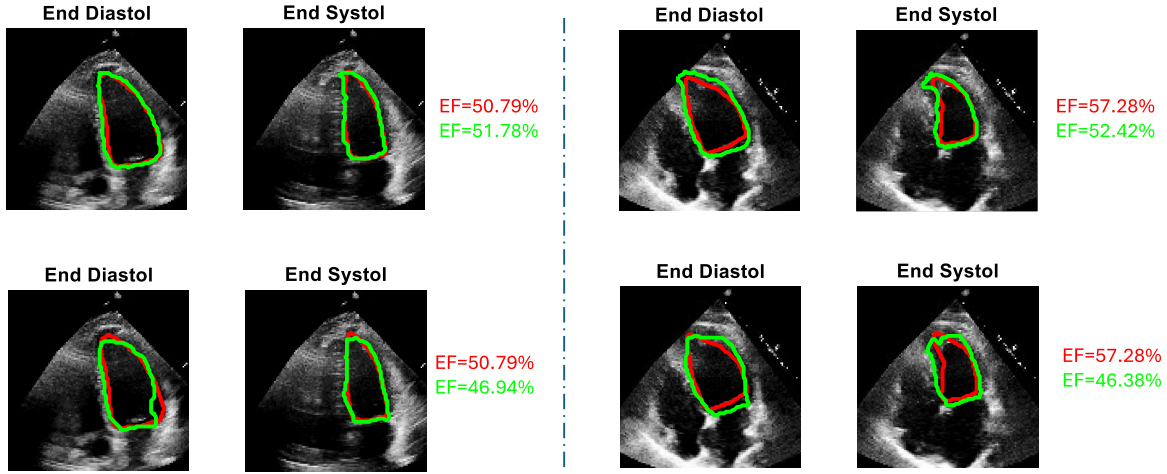


Fig. 9. Examples of left ventricular segmentation from the EchoNet dataset using a model trained on EchoNet (first row) and CAMUS (second row). The green lines indicate the boundaries of the segmented mask using the proposed model, while the red lines indicate the ground-truth boundaries.

TABLE V
COMPUTATIONAL ANALYSIS OF THE COMPARED MODELS

Method	Backbone Model	No. Parameters	Training Time (Minutes)
UNET	CNN	22M	58
UNET	Resnet 34	36M	165
UNET	Resnet 50	51M	141
FPNet	Resnet 34	31M	219
FPNet	Resnet 50	38M	176
Pix2Pix GAN (ours)-Generator	UNET/DCNN	54.43M	138
Pix2Pix GAN (ours)-Discriminator	Patch GAN	6.97 M	138
Pix2Pix GAN (ours)-Entire Training	Generator and Discriminator	61.4 M	138

loss functions. However, in comparison with other models, it is worth noticing that UNET (the CNN backbone) utilized fewer training parameters and computational time, but its results on the segmented cardiac regions were not promising (see Fig. 5). On the other hand, even though our approach required more computational time because of its compact representation of UNET and PatchGAN, it demonstrated more accurate and precise segmentation on all three cardiac regions, outperforming the state-of-the-art models.

F. Generalization Capability of the Proposed Model

Assessing the generalization capabilities of our proposed segmentation model is crucial. We utilized the EchoNet dataset, which consists of 1277 videos in the test set [47]. The dataset is publicly available online on the EchoNet website (see <https://aimi.stanford.edu/echonet-dynamic-cardiac-ultrasoundlink>). Our evaluation involved two distinct strategies: first, we trained the model using a small subset of training data, specifically 1500 end-diastole (ED) and 1500 end-systole (ES) frames from 1500 videos in the training set and subsequently tested it on the complete EchoNet test set. Second, we applied the model trained on the CAMUS dataset to the entire EchoNet test set to assess its performance on datasets with diverse acquisition settings not encountered during training. The results, detailed in Table VI and Fig. 10 and exemplified by two segmentation examples in Fig. 9, underscore the model's successful generalization to the EchoNet dataset even when trained on a limited subset

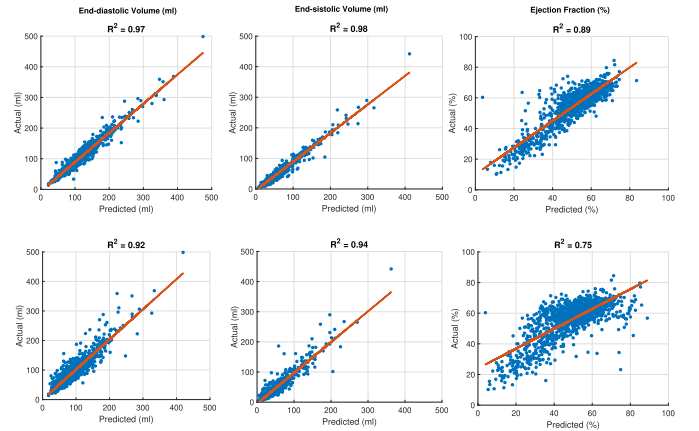


Fig. 10. R^2 correlation coefficient for the LV_{EDV} (first column), LV_{ESV} (second column), and EF (third column). The results are extracted for the EchoNet dataset using a model trained on the EchoNet (first row) and CAMUS (second row) datasets.

TABLE VI
QUANTITATIVE RESULTS ON THE ECHONET DATASET

	Mean Dice	correlation			MAE		
		LV_{EDV}	LV_{ESV}	LV_{EF}	LV_{EDV} (ml)	LV_{ESV} (ml)	LV_{EF} (%)
Trained on EchoNet	0.89	0.97	0.98	0.89	12.716	8.613	4.87
Trained on CAMUS	0.79	0.92	0.94	0.75	12.588	8.602	9.02

of the training data. Moreover, applying the model trained on CAMUS data to the EchoNet test set demonstrates promising generalizability in terms of geometric and clinical parameter estimation.

V. CONCLUSION

This study addressed a notable gap in left ventricular echocardiography ultrasound data analysis by proposing a robust segmentation approach to enhance the accuracy of clinical and geometrical parameters. The introduced strategy focused on refining the segmentation performance of the LV_{endo} , myocardium, and atrium through a pixelwise segmentation technique based on the Pix2Pix GAN model.

The Pix2Pix GAN framework leverages the UNET and PatchGAN models as backbone components, proving exceptionally valuable for the accurate segmentation of 2-D echocardiography data. We applied our proposed technique to CAMUS 2-D echocardiography data. Through an extensive series of experiments, we conducted a comprehensive analysis to compare the state-of-the-art methodologies, including ACNN [37], UNET [38], CLAS [40], and GUDU [39] networks. Overall, our experimentation demonstrated that the proposed Pix2Pix GAN approach outperformed the existing methods, with an effective balance between clinical and geometric parameters. Moreover, our experimental results revealed substantial enhancements in geometric parameters, such as the Dice coefficient, for the segmentation of the LV. Nevertheless, our method consistently achieved the highest segmentation results when assessing clinical parameters, particularly in relation to the analysis of EF for the LV_{endo}. In summary, this research underscores the promise of the Pix2Pix GAN-based approach in tackling the complexities linked to left ventricular echocardiography ultrasound data analysis. It contributes to the enhancement of clinical parameter assessment and geometric parameter evaluation. Furthermore, this methodology holds the potential to extend its utility to various other medical imaging segmentation tasks.

In future studies, we aim to shift our focus toward video-level, real-time segmentation through the utilization of a 3-D GAN architecture.

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